

CSCI 1302 Final Exam - Study Topics

This guide outlines the major topics covered throughout the semester. The final exam is comprehensive but will place additional emphasis on material covered since Exam 3, particularly Abstract Classes/Interfaces, JavaFX, Event Handling, Binary I/O, and Recursion. Reviewing previous exams, labs (especially Labs 11 & 12), and lecture notes is highly recommended.

Objects and Classes (Review)

Defining classes, constructors (default, convenience)

Object instantiation (new keyword), reference variables

Instance vs. static members (variables and methods)

Visibility modifiers (public, private, protected, default)

Data field encapsulation (private data, public accessors/mutators)

Passing objects to methods, returning objects from methods

Arrays of objects

Immutable objects and classes

Scope of variables

this keyword (accessing members, invoking constructors)

Object-Oriented Thinking (Review)

Class abstraction and encapsulation

Class relationships (association, aggregation, composition)

Representing relationships in UML (arrows, diamonds)

Wrapper classes for primitive types (e.g., Integer, Double)

Autoboxing/unboxing

The String, StringBuilder, and StringBuffer classes

Inheritance and Polymorphism (Review)

Superclasses and subclasses (extends keyword)

super keyword (calling superclass constructors and methods)

Method overriding

Polymorphism

Dynamic binding

Casting objects (instanceof operator, upcasting, downcasting)

Overriding the equals method (for object comparison)

ArrayList class and common methods

protected visibility

final keyword (for classes, methods, variables)

Exception Handling and Text I/O (Review)

Exception handling (try, catch, finally)

Exception hierarchy (Exception, RuntimeException, IOException)

Checked vs. Unchecked exceptions

Throwing exceptions (throw keyword)

Rethrowing exceptions, chained exceptions

Custom exception classes
File class (representing files/directories)
Text File I/O: Scanner (input), PrintWriter (output), FileReader, FileWriter
try-with-resources statement for automatic resource closing

Abstract Classes and Interfaces (Ch 13 & Exam 3 Emphasis)

Abstract classes (abstract keyword)

Abstract methods (no implementation)
Purpose and use of abstract classes (commonality, templates)
Cannot instantiate abstract classes directly
Interfaces (interface keyword)
Interface characteristics (constants, abstract methods, default methods, static methods)
Implementing interfaces (implements keyword)
Marker interfaces (Serializable, Cloneable)
Comparable interface (compareTo method)
Cloneable interface (shallow vs. deep copy concepts)
Relationship between interfaces, abstract classes, and polymorphism

JavaFX Basics (Ch 14 & Exam 3 Emphasis)

JavaFX application structure (Application class, start method)
Stage, Scene, Node hierarchy

Common UI controls (Label, Button, TextField, CheckBox, RadioButton, TextArea, ComboBox, ListView, ScrollBar, Slider, etc.)
Layout Panes (Pane, HBox, VBox, BorderPane, GridPane, StackPane, FlowPane, etc.)
Shapes (Line, Rectangle, Circle, Ellipse, etc.)

Properties and Property Binding

Common Node properties/methods (style, layoutX/Y, translate, rotate, etc.)
Color and Font classes
Image and ImageView classes

Event-Driven Programming and Animations (Ch 15 & Exam 3 Emphasis)

Event-driven programming paradigm
Events and Event Sources (e.g., Button)

Event Handlers/Listeners
Registering handlers (setOnAction, etc.)
Handling events (ActionEvent, MouseEvent, KeyEvent)
Inner classes (including Anonymous Inner Classes) for handlers
Lambda expressions for concise handlers
Observable objects
Animations (Timeline, PathTransition, FadeTransition, etc.)
Multimedia (Media, MediaPlayer, MediaView for video/audio)

Binary I/O (Ch 17 & Final Exam Emphasis)

Text files vs. Binary files (representation, efficiency)

Java I/O class hierarchy (InputStream, OutputStream)

Byte streams: FileInputStream, FileOutputStream

Filtering streams (FilterInputStream, FilterOutputStream)

Data streams: DataInputStream, DataOutputStream (reading/writing primitive types and UTF strings)

Methods: readInt(), writeInt(), readDouble(), writeDouble(), readBoolean(), writeBoolean(), readUTF(), writeUTF(), readChar(), writeChars()

Importance of reading data in the same order/format as written

Buffering streams: BufferedInputStream, BufferedOutputStream (performance benefits)

End-of-File detection (EOFException with data/object streams, -1 with basic byte streams)

Object streams: ObjectInputStream, ObjectOutputStream

Writing/reading objects (writeObject(), readObject())

Requirement: Objects must implement Serializable interface

Serializing arrays and collections

transient keyword (excluding fields from serialization)

RandomAccessFile

Reading and writing to the same file

File pointer (seek(), getFilePointer(), length())

Modes ("r", "rw")

Working with fixed-length records

Recursion (Ch 18 & Final Exam Emphasis)

Concept of recursion (method calling itself)

Anatomy of a recursive method:

Base case(s) (stopping condition)

Recursive step(s) (reducing the problem towards the base case)

Tracing recursive calls (call stack concept)

Examples:

Factorial

Fibonacci numbers

Printing messages/patterns

Palindrome checking

Greatest Common Divisor (GCD)

Array processing (sum, search, sort)

Directory traversal/size calculation

Tower of Hanoi

Fractals (e.g., Sierpinski triangle)

Recursive helper methods (simplifying initial call, managing state)

Recursion vs. Iteration (trade-offs in readability, performance, memory usage)

Potential for `StackOverflowError` with deep or infinite recursion
Tail recursion (concept, potential compiler optimizations)